

Hard Disk Drives

CS311 Operating System, Fall 2020

HDD

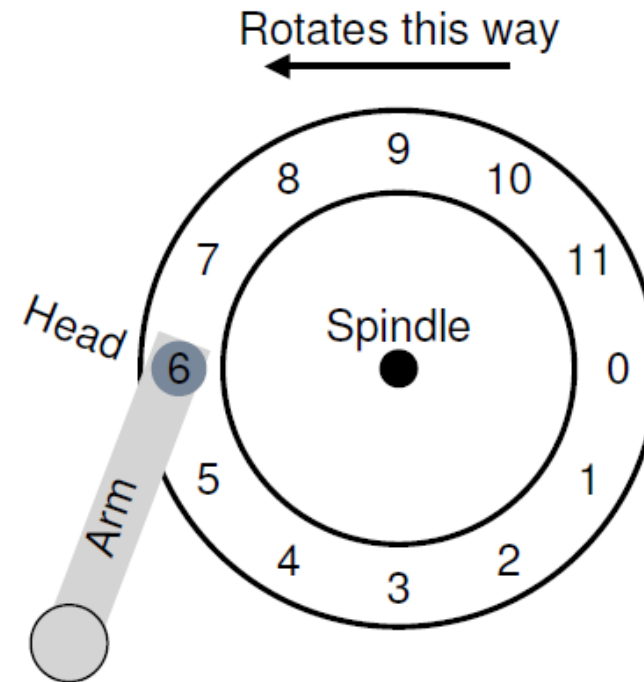
- An I/O device
- Used for persistent storage
- Consists of sectors (512 byte each) numbered 0 to $n-1$
- Writes to a sector are atomic
- Bigger read/writes are possible but not atomic
- Accessing blocks near to each other is faster than accessing blocks far away from each other

HDD – physical geometry

- Platter
 - Made of metal (aluminium) coated with magnetic material on both sides
 - Data is written by inducing magnetic changes
 - A disk may have one or more platters
- Surface: a side of a platter
- Spindle: binds together multiple platters
- Speed: Revolutions per minute (RPM)
- Tracks: concentric circle made up of sectors
- A Disk Head: is used to read/write data
- A Disk Arm: is used to move the head, position it over the right track

HDD – working

- Rotational Delay
 - To read track 0?
 - Max rotational delay?

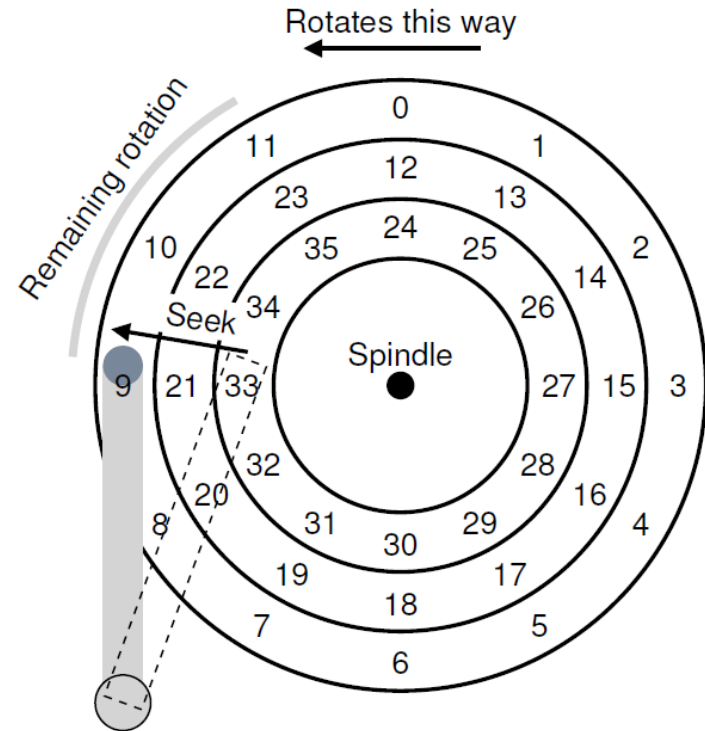
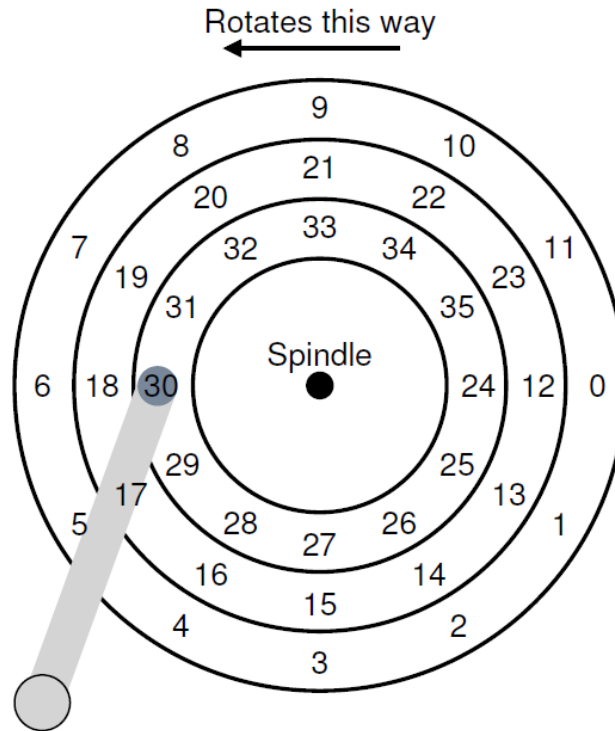


HDD – working: seek operation

- Seek phases = acceleration + coasting + deceleration + settling

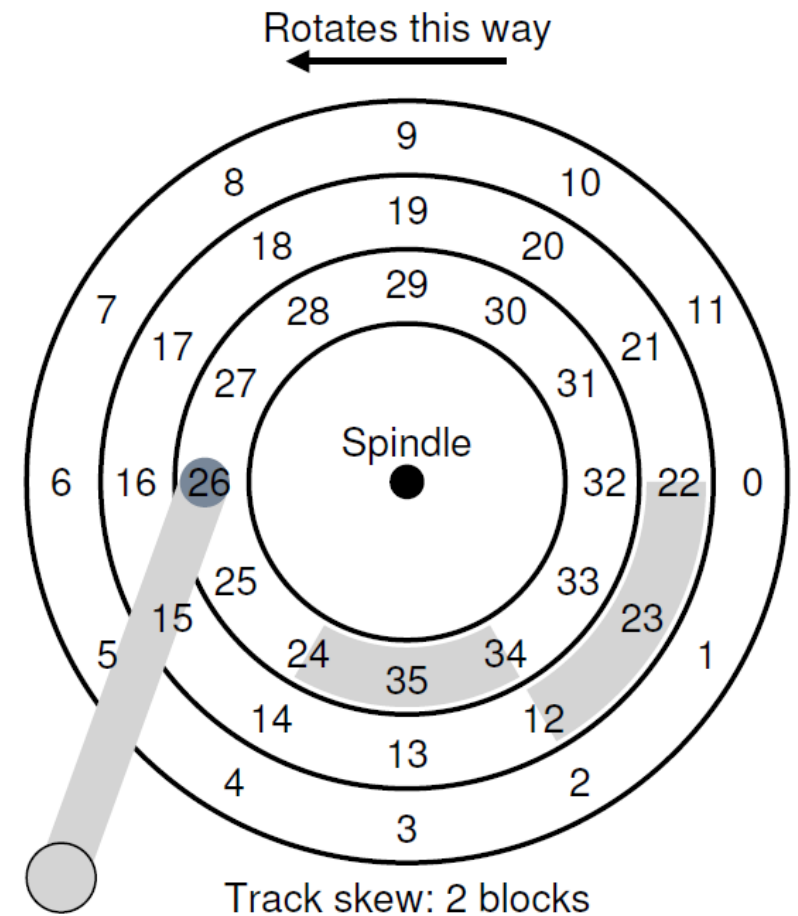
- Seek Time

- I/O time =
seek time +
rotate time +
transfer



Track skew

- To facilitate transfers crossing track boundaries



Track buffer

- Works as a Cache
 - Reads and stores multiple consecutive sectors for future use
- Writing can have two strategies:
 - Write back
 - Write through

HDD specs

	Cheetah 15K.5	Barracuda
Capacity	300 GB	1 TB
RPM	15,000	7,200
Average Seek	4 ms	9 ms
Max Transfer	125 MB/s	105 MB/s
Platters	4	4
Cache	16 MB	16/32 MB
Connects via	SCSI	SATA

I/O Time

$$T_{I/O} = T_{seek} + T_{rotation} + T_{transfer}$$

$$R_{I/O} = \frac{Size_{Transfer}}{T_{I/O}}$$

On the Cheetah:

$$T_{seek} = 4 \text{ ms}, T_{rotation} = 2 \text{ ms}, T_{transfer} = 30 \text{ microsecs}$$

T I/O = 6 ms, R I/O = 0.66 MB/sec

For barracuda: T I/O = 13.2 ms, R I/O = 0.31 MB/sec

Sequential Transfer (100MB)

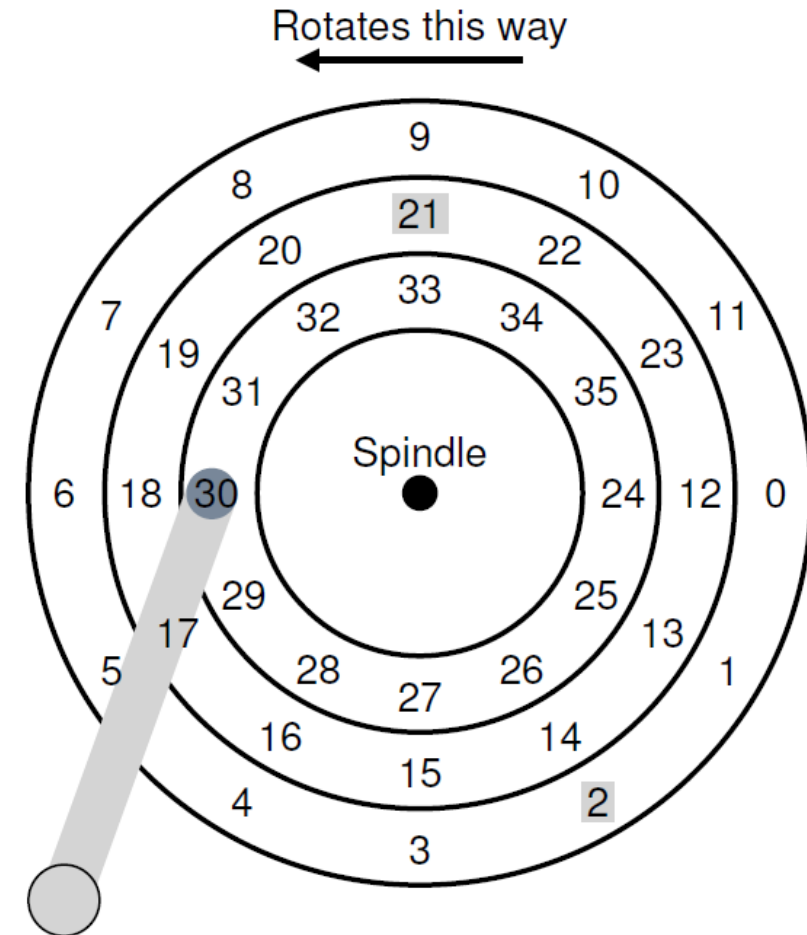
- Cheeta T I/O = 800 ms
- Barracuda T I/O = 950 ms

	Cheetah	Barracuda
$R_{I/O}$ Random	0.66 MB/s	0.31 MB/s
$R_{I/O}$ Sequential	125 MB/s	105 MB/s

- Conclusion: Drive performance varies heavily whether we are doing random small transfers or big sequential transfers !!

Disk scheduling

- Multiple I/O requests of varying characteristics
- How can we perform them efficiently?
- Disk scheduler tries to schedule them to reduce average delay.
- Length of job is known ??
- SJF == SSTF == NBF
- Starvation?

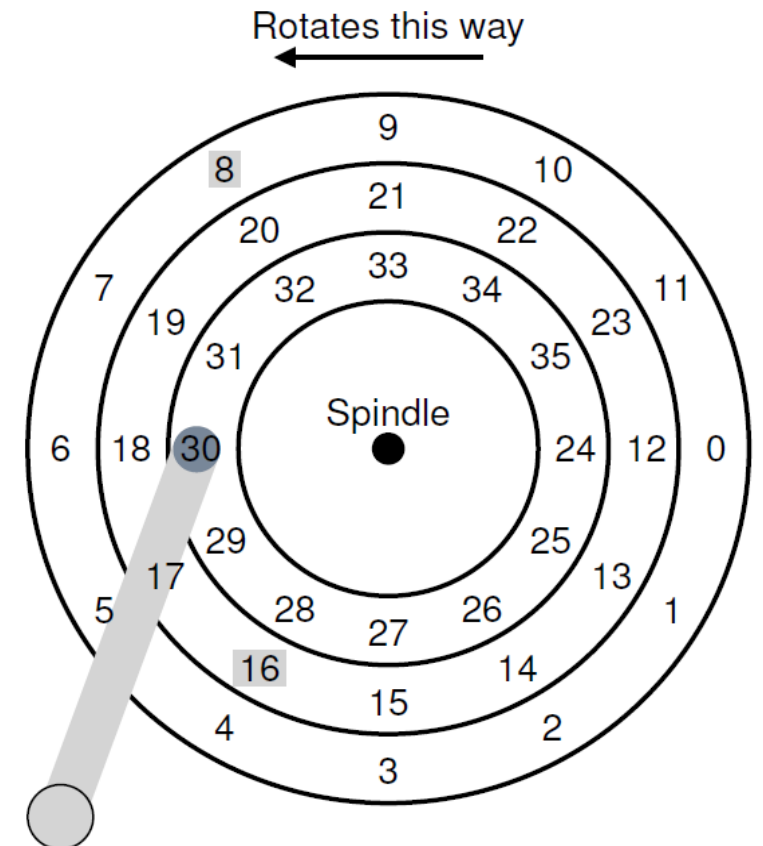


SSTF can lead to starvation

- SCAN algorithm remedies this
- It moves the head in a sweep (from inner to outer tracks and vice versa)
- The next request served will be the closest one but not on the same track!
- Variations: FSCAN, CSCAN
- Also called the Elevator algorithm.

SPTF (shortest positioning time first)

- Which sector to read next: 16 or 8?
- SSTF and SCAN do not take into account the time to rotate.
- Problem: often OS does not know the internal details of the disk!



Disk Scheduling – issues

- Where is the scheduling performed?
 - OS
 - HDD
- I/O merging
 - Imagine 3 requests to read from sectors 33, 8, 34 respectively
- How long the system should wait before issuing an I/O request to disk?
 - Immediately (work conserving), or
 - After some time (non work conserving)